

Sri Lanka Institute of Information Technology



Lab Submission
Lab - 05

IT25102546

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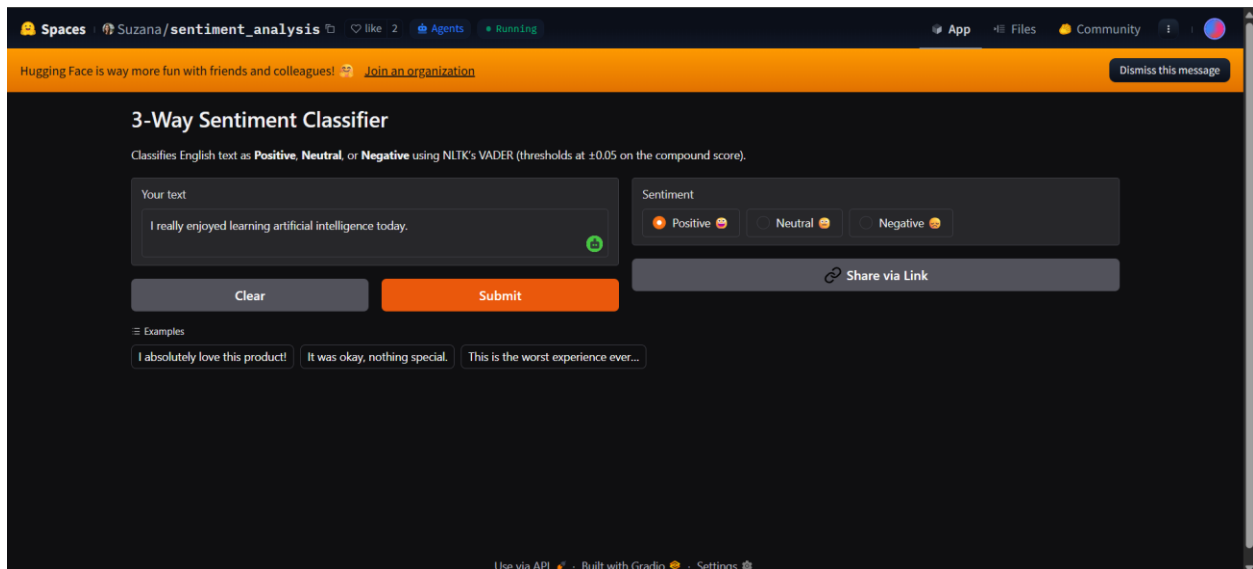
Artificial Intelligence and Machine Learning | IT2011

B.Sc. (Hons) in Information Technology

Exercise 5

- **Selected Hugging Face Space: 3-Way Sentiment Classifier**
- **Purpose:** This Space classifies English text as **Positive**, **Neutral**, or **Negative**.
- **Input:** I really enjoyed learning artificial intelligence today.
- **Generated Output:** Positive

Screenshot:



Selected Model: distilbert-base-uncased-finetuned-sst-2-english

Developer: Hugging Face

Task: Text Classification / Sentiment Analysis

Language: English

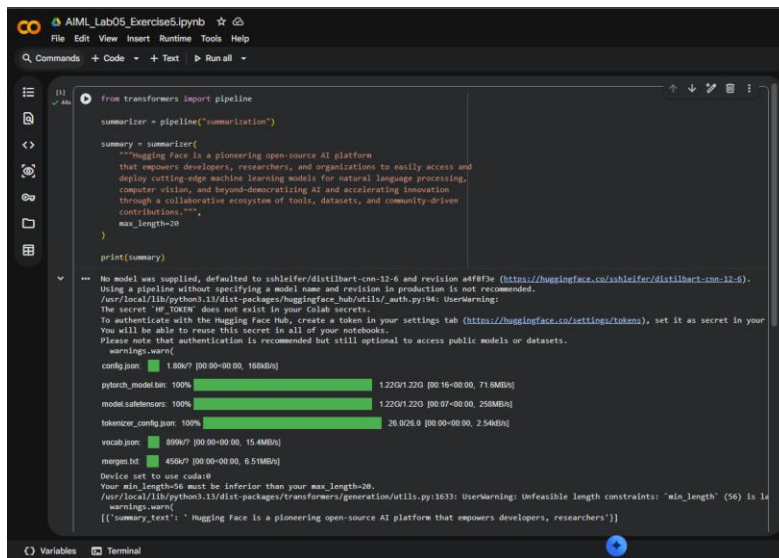
License: Apache-2.0

Model Size: 67 million parameters

Model Description: This is a DistilBERT model fine-tuned on the SST-2 dataset. It can classify English text as positive or negative sentiment.

Reason for Selection: I selected this model because it is easy to use and can identify whether an English sentence expresses positive or negative sentiment.

Step 9 – General Summarization



```
[1] from transformers import pipeline

summarizer = pipeline("summarization")

summary = summarizer(
    """Hugging Face is a pioneering open-source AI platform
    that empowers developers, researchers, and organizations to easily access and
    deploy cutting-edge machine learning models for natural language processing,
    computer vision, and beyond—democratizing AI and accelerating innovation
    through a collaborative ecosystem of tools, datasets, and community-driven
    contributions.""",
    max_length=20
)

print(summary)
```

-- No model was supplied, defaulted to sshleifer/distilbert-cnn-12-6 and revision 4d083b (https://huggingface.co/sshleifer/distilbert-cnn-12-6).
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your GitLab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json: 1.80K/7 [00:00<00:00, 168KB/s]
pytorch_model.bin: 100% 1.22G/1.22G [00:16<00:00, 71.6MB/s]
model.safetensors: 100% 1.22G/1.22G [00:07<00:00, 258MB/s]
tokenizer_config.json: 100% 25.0/25.0 [00:00<00:00, 2.54KB/s]
vocab.json: 899K/7 [00:00<00:00, 15.4MB/s]
merges.txt: 458K/7 [00:00<00:00, 6.51MB/s]
Device set to use cuda:0
Your min_length=56 must be inferior than your max_length=20.
/usr/local/lib/python3.11/dist-packages/transformers/generation/utils.py:1633: UserWarning: Unfeasible length constraints: 'min_length' (56) is la
warnings.warn(
[{'summary_text': 'Hugging Face is a pioneering open-source AI platform that empowers developers, researchers'}]]

Step 11 – KoBART Summarization Model

```
from transformers import pipeline

# Use a known summarization model
summarizer = pipeline(
    "summarization",
    model="EbanLee/kobart-summary-v3"
)

# Input text to summarize
text = """Hugging Face is a pioneering open-source AI platform
that empowers developers, researchers, and organizations to easily access and
deploy cutting-edge machine learning models for natural language processing,
computer vision, and beyond-democratizing AI and accelerating innovation
through a collaborative ecosystem of tools, datasets, and community-driven
contributions."""

# Generate summary
summary = summarizer(
    text,
    max_length=20,
    min_length=5,
    do_sample=False
)

print(summary)
```

... config.json: 1.68k/? [00:00<00:00, 96.3kB/s]
You passed 'num_labels=3' which is incompatible to the 'id2label' map of length '2'.
You passed 'num_labels=3' which is incompatible to the 'id2label' map of length '2'.
model.safetensors: 100% 496M/496M [00:05<00:00, 103MB/s]
generation_config.json: 100% 191/191 [00:00<00:00, 17.8kB/s]
tokenizer_config.json: 39.5k/? [00:00<00:00, 3.56MB/s]
tokenizer.json: 1.05M/? [00:00<00:00, 37.7MB/s]
special_tokens_map.json: 100% 692/692 [00:00<00:00, 33.4kB/s]
Device set to use cuda:0
Both 'max_new_tokens' (=256) and 'max_length' (=20) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the documentation for more information
[{'summary_text': 'Hugging Face is a pioneering open-source AI platform\nthat empowers developers, researchers, and organizations to easily access and deploy cutting-e

Step 13 – Selected NLP Model: Sentiment Analysis

Selected Model: distilbert/distilbert-base-uncased-finetuned-sst-2-english

Task: Sentiment Analysis

Reason for Selection: I selected this model because it is easy to use and can identify whether an English sentence expresses positive or negative sentiment.

Input: I really enjoyed learning artificial intelligence today.

Output: [{'label': 'POSITIVE', 'score': 0.9996747970581055}]

Interpretation: The model classified the sentence as positive with approximately 99.97% confidence.

```
from transformers import pipeline

# Load the selected sentiment analysis model
classifier = pipeline(
    "sentiment-analysis",
    model="distilbert/distilbert-base-uncased-finetuned-sst-2-english"
)

# Input text
text = "I really enjoyed learning artificial intelligence today."

# Analyze the sentiment
result = classifier(text)

# Display the result
print(result)
```

... config.json: 100% 629/629 [00:00<00:00, 55.1kB/s]
model.safetensors: 100% 268M/268M [00:05<00:00, 46.7MB/s]
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.48kB/s]
vocab.txt: 232k/? [00:00<00:00, 17.5MB/s]
Device set to use cuda:0
[{'label': 'POSITIVE', 'score': 0.9996747970581055}]